ONE-PAGER · BRIEF 12 · ACCESS · AFFORDABILITY · AVAILABILITY · V2

Relocatable clean power for a hospital on the move

Team Secure AI Grid · Pavan Manjunath, Thomas Prüfer | **Solution** GridRescue AI | **Worked example** National Orthopaedic Hospital Enugu

About half of the hospitals in sub-Saharan Africa have reliable electricity [24], most firms in the region report power outages [26], and Africa holds about 60 % of the world's best solar resources but about 1 % of its installed solar capacity [25]. In Enugu: about 6 h of grid a day for 1,550 kW [1], diesel ₦3,277/L, up 86 % (NBS national average [4]), and a move in about five years [11]. **GridRescue AI**: vendor-neutral software, map pin → tested, costed, relocatable plan and ESG report, in minutes.

THE THREE A'S IN ENUGU · ENGINE RESULTS ARE MODELLED, ±30 % [13]

	WHAT GRIDRESCUE AI DOES · BEFORE → AFTER (MODELLED OUTAGES)
ACCESS	Any site read from open data in seconds: 9 substations within 50 km, grid class D, 5.48 peak-sun hours, Band A ₦160/kWh [5]; the 303 kW critical load first. **Theatre-hours a year without grid: 4,992 (40 %) → 0.**
AFFORDABILITY	1,670 kWp solar, 5,020 kWh containerised energy system, 500 kVA UPS; capex ₦3.85 bn. Bought outright: diesel savings repay it and the move in 1.6 years. ESCO: ₦0 upfront. **Energy costs, year one: ₦3,764.5 M → ₦1,588.9 M.**
AVAILABILITY	Online UPS on theatres (0 ms switchover, modelled), solar by day, energy containers by night, diesel last; 4/4 stress tests; 88 % reused in the move. **Switchovers about 1,460 a year → 0; CO_2 2,795 t → 549 t a year.**

Today = grid + modelled diesel; with the system = ₦69.1 M/month ESCO fee (the hospital's cost, not our revenue) + remaining diesel and grid; saves from month one if the real diesel bill exceeds ₦117.9 M/month.

BUILT DURING THE GLOBAL ENERGY HACKATHON 2026

What we built, in short: a front-end GUI grid mapper (sun · wind · water · grid lenses and the new-site lens) → a basic ML classification of the green energy system (Random Forest picks the energy system; the hourly simulation verifies it) → a location-based ESG report, downloadable for any site.

1 GRID MAPPER	2 RELOCATION PLAN	3 3D SIMULATION + ML	4 ESG REPORT
Sun, wind, water, grid lenses; substations, supply hours, tariff; new-site lens.	Brief 12's five questions: theatres first, movable system, move plan, money, funders.	Hour by hour, four stress tests; Random Forest pick, 80.4 % on held-out synthetic sites.	Scope 1 / 2, E·S·G lines, before → after KPIs, before anything is bought.

ESG report download: Relocation plan › Impact tab (Download ESG report), Simulate page (ESG report button) and Report page; it works for any pinned location (tested: Enugu, Kano, Nairobi, Accra) and gives Scope 1 · Scope 2 (GHG Protocol), E·S·G report lines and carbon value at the buyer's price — a board-ready, business-standard report (modelled, ±30 %).

Pre-existing base: to be confirmed. **Five outage challenges** · built: grid interruption (12 h), grid collapse (24 h), worst solar month, diesel last (18 h) · planned: cyber / operational disruption as a stress scenario only.

MARKET RESEARCH · TOOLS, WHAT IS DIFFERENT, WHAT SCALES

Tool	Grid mapping	Hourly outage sim.	ML energy-system pick	Relocation plan	Financing / cost	ESG report	No expert needed
⚡ REopt (NREL)	✗	✓	✗	✗	◐	✗	✗
⚙ HOMER Pro (UL Solutions)	✗	✓	✗	✗	✓	✗	✗
✚ HOMER Powering Health (World Bank/ESMAP)	✗	✗	✗	✗	✓	✗	–
$ Odyssey Energy Solutions	✗	✗	✗	✗	✓	✗	–
⌖ MapYourGrid / OpenStreetMap	✓	✗	✗	✗	✗	✗	–
★ GridRescue AI	✓	✓	✓	✓	✓	✓	✓

✓ *yes* ◐ *partly* ✗ *no* – *not stated in our sources* *Sources: the tools' websites (refs in the POC summary).*

What scales · any pin on the map · a new country adds its tariff table and local prices · from hospitals to every critical facility (data centres, water, telecom) · further outage types (weather, transmission, cyber / operational as stress scenarios).

CUSTOMER, USER, PAYER · BUSINESS MODEL · ASK (INVESTOR VIEW)

First customer and user · the Enugu hospital (modelled pilot); federal and teaching hospitals on weak grids first; users: facilities teams.
Three money stories, never mixed · (a) hospital saves diesel minus fee; (b) an ESCO finances and moves the hardware; (c) hospital or health network buys our software: US$ 2,000 assessment + US$ 3,000/site/year licence, enterprise and white-label (proposed).
The ask · US$ 150,000 seed, two US$ 75,000 tranches (second after hospital-data validation and a pilot partner); lean US$ 75,000, ~6 months. Use (US$ k): engineering 70 · validation 20 · security 12 · cloud/data 8 · pilots 15 · acquisition 10 · legal/IP 5 · contingency 10. **Roadmap** · 0–3 m validate · 3–9 two paid pilots · 9–12 release, 10 paying sites · 12–18 utilities, new countries.

JUDGING FIT [11] Impact 25 %: theatre-hours without grid 4,992 → 0, 2,246 t CO_2 a year (modelled) · Innovation 25 %: relocatable workflow · Sustainability 25 %: 88 % reused · Feasibility 15 %: 4/4 stress tests · Scalability 10 %: global open data.

OUR TEAM SUBMITS Demo + pitch videos (< 5 min), deck, one-pager, POC summary, canvas, financial model, disclosure.

GridRescue AI turns energy planning into investment-ready decisions. We start with hospitals, where power is life, and grow to every critical facility. Map the grid. Keep the theatres on. Move with the hospital.

KEY REFERENCES

[1] National Orthopaedic Hospital Enugu, head engineer, problem statement, 2026 (in Problem Brief 12).
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[5] Nairametrics, "Enugu slashes electricity tariff for Band A customers to N160/kWh effective August 1," 2025. [Online]. Available: https://nairametrics.com/2025/07/21/enugu-slashes-electricity-tariff-for-band-a-customers-to-n160-kwh-effective-august-1/
[11] Global Energy Hackathon, "Problem Brief 12: Relocatable solar for a hospital on the move," hackathon.anneru.com.
[13] GridRescue AI engine output for the Enugu worked example (modelled, ±30 %; to be confirmed with the hospital's load data).
[24] WHO, World Bank, IRENA and SEforALL, Energizing health, 2023. [Online]. Available: https://www.who.int/publications/i/item/9789240066960
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